BIO302 FINALTERM SUBJECTIVE MEGA FILE

ALL DATA FROM PAST PAPERS (2018-2022)

GROUP: BIOTECH BRAINY BUNCH

ADMINS: MEHWISH MUGHAL & BELLA

CONTACT: mehwishshabirhussain07@gmail.com

SHORT QUESTIONS

1) **Two use of nitrogen mustard gas?**

Mustard gas is a vesicant warfare agent synthesized by Fredrick Guthrie in 1860. It was widely used as a weapon during WWI by both sides of the conflict with particularly harmful and deadly effects**.** It was responsible for 1,205,655 non-fatal casualties and 91,198 deaths.

2) **What are psi and phi values in a helix structure and mention value?**

The phi angle is the angle between the alpha carbon atom and the nitrogen while the psi angle is the angle between the alpha carbon and the carbon of the carbonyl group. Typical value are Φ = -140 degrees and Ψ = 130 degrees. In contrast, alpha helical residues have both negative φ and ψ angles. A section of polypeptides residue with in the beta-conformation is referred to as a beta strand can associate by main chain hydrogen bonding interactions to forma sheet.

3) **How many classes of splicing found in cell?**

In cells, two type of splicing is found including alternative splicing and constitutive splicing. There are three kinds of self-splicing introns, Group I, Group II and Group III. Group I and II introns perform splicing similar to the spliceosome without requiring any protein. The similarity suggests that Group I and II introns may be evolutionarily related to the spliceosome.

4) **What are the two RNA polymerase present in the eukaryotic other than RNA pol II and how the different from RNa pol II?**

Transcription in eukaryotes involves one of the three types of polymerases, depending on the gene being transcribed. RNA polymerase II transcribes all of the protein-coding genes, whereas RNA polymerase I transcribes rRNA genes, and RNA polymerase III transcribes rRNA, tRNA, and small nuclear RNA genes.

5) Define nuclear Scaffold, bacterial core and enzyme?

Nucleus Scaffold: The activity of type II topoisomerases is also critical to the segregation of large linear molecules. Although there is no inherent topological linkage after the replication of a linear molecule, the large sized chromosomes necessitates the intricate folding of the DNA into loops which are attached to a protein scaffold. These attachments lead to many of the same problems that circular chromosomes have after replication. This process occurs in nucleus, that's why it is known as nucleus scaffold.

Bacterial Core: The bacterial core genes are defined as single-copy, homologous genes that are present in most of the known bacterial species.

Enzyme: An enzyme is a biological catalyst and is almost always a protein. It speeds up the rate of a specific chemical reaction in the cell. The enzyme is not destroyed during the reaction and is used over and over.

6) What will happen to cellular machinery and function if pyrimidine dimer is not removed?

These dimers distort the DNA structure and results in errors during DNA replication. They form an extra phosphodiester bond between them. They prevent the transcription of the DNA into RNA. If this damage does not repair, it will cause severe diseases such as Xeroderma pigmentosum and Cockayne Syndrome. Affected individuals have extreme sensitivity to the sunlight and develop skin cancers due to thymine dimer passing on generations to generations.

7) Mismatch repair system ?

Essential components of the MMR-system are MutS, MutL, MutH and Uvr.DNA mismatch repair functions to correct replication errors in newly synthesized DNA and to prevent recombination between related, but not identical (homologous), DNA sequences. The mechanism of mismatch repair is best understood in Escherichia coli.

8) Names of types of Mutations?

There are three types of DNA mutations:

- base substitutions
- deletions
- insertions.

DO NOT COPY

9) Write types of splicing?

Two different modes of splicing have been defined, that is: constitutive splicing and alternative splicing.

10) Template primer junction?

The primer is a short piece of RNA or DNA that is complementary to the 3' end of the template DNA and which provides a 3'-OH for DNA polymerase to add new nucleotides to. DNA polymerase cannot begin synthesis of a new strand of DNA without a primer.

11) Secondary structure of RNA?

Nucleus acid secondary structure is the base pairing interactions within a single nucleic acid polymer or between two polymers. I can be represented as a list of bases which are paired in a nucleic acid molecule. At a fundamental level, RNA secondary structure consists of nucleotides that are in one of two states, paired or unpaired, where pairing includes all base-base interactions. In general, most base pairings are adjacent and antiparallel with other base pairings to form secondary structure helices.

12) Explain termination sequence?

Two termination mechanisms are well known:

Intrinsic termination: (also called Rho-independent transcription termination) involves terminator sequences within the RNA that signal the RNA polymerase to stop. The terminator sequence is usually a palindromic sequence that forms a stem-loop hairpin structure that leads to the dissociation of the RNAP from the DNA template.

Rho-dependent termination: uses a termination factor called ρ factor (rho factor) which is a protein to stop RNA synthesis at specific sites. This protein binds at a rho utilisation site on the nascent RNA strand and runs along the mRNA towards the RNAP. A stem loop structure upstream of the terminator region pauses the RNAP, when ρ-factor reaches the RNAP, it causes RNAP to dissociate from the DNA, terminating transcription

OR

Once the polymerase has transcribed the length of the gene (or genes), it must stop and release the RNA product (as well as dissociating from the DNA itself). This step is called termination.

13) Write function of RNA ploymerase?

RNA polymerase is an enzyme that is responsible for copying a DNA sequence into an RNA sequence, duyring the process of transcription.In bacteria, RNA polymerase exists in two states. One form, known as the core enzyme, can catalyze RNA synthesis but is unable to bind to

DO NOT COPY

promoter targets in DNA. The second form of RNA polymerase, the holoenzyme, is capable of both RNA synthesis and promoter recognition.

14) How RNA ploymerase II differ from I and III?

RNA Polymerase I is an enzyme that transcribes ribosomal RNAs. RNA Polymerase II is an enzyme that transcribes precursors of mRNAs. RNA Polymerase III is an enzyme that transcribes tRNAs. It transcribes all rRNAs except the 5S rRNA component. These three polymerases are used in three different specified mechanisms.

15) Define splisome?

A spliceosome is a large and complex molecular machine found primarily within the splicing speckles of the cell nucleus of eukaryotic cells. The spliceosome is assembled from snRNAs and protein complexes. The spliceosome removes introns from a transcribed pre-mRNA, a kind of primary transcript. This process is generally referred to as splicing. Only eukaryotes have spliceosomes and metazoans have a second spliceosome, the minor spliceosome.

16) Define major nd minor groove?

the major groove is wider than the minor groove. These grooves allow proteins to bind to and recognize DNA sequences from the outside of the helix. The grooves expose the edges of each base pair located inside the helix, which allows proteins to chemically recognize specific DNA sequences.

17) What is diester bond?

A bond between a two sugar groups and a phosphate group; such bonds form the sugar-phosphate-sugar backbone of DNA and RNA. A diester bond (between phosphoric acid and two sugar molecules) linking two nucleotides together to form the nucleotide polymers DNA and RNA.

18) Differentiate between endonucleases and exonucleases causing agent?

DNA exonucleases cleave the sequence from either end, whereas DNA endonucleases cleave the sequence within or adjacent to the sequence.

19) What is elongation in RNA polymerase?

Basically, elongation is the stage when the RNA strand gets longer, thanks to the addition of new nucleotides. During elongation, RNA polymerase "walks" along one strand of DNA, known as the template strand, in the 3' to 5' direction.

20) Define Torpedo model?

DO NOT COPY

The torpedo model suggests that a 5' to 3' exonuclease degrades the second RNA as it emerges from the elongation complex. Polymerase is released as the highly processive exonuclease overtakes it.

OR

An enzyme that degrades the second RNA as it emerges from the polymerase has been identified, and this enzyme may itself trigger termination. This is called the torpedo model of termination.

21) mRNA has short life span. Write experiment?

Short lifespan: Messenger RNA is much smaller than DNA and is far less stable. It has a very short half-life (from seconds to minutes or hours) compared to that of nuclear DNA (years). Because it has a short half-life, the purine and pyrimidine bases that are used to make mRNA must be continually resynthesized.

Experiment: mRNA half-life can be determined using Northern blot analysis or quantitative RT-PCR (qRT-PCR) analysis. Depending on the amount of RNA required, one should choose the number of cells to be seeded.

22) Describe Group 1 intron splicing?

Many of the group I and group II introns are self-splicing in that no additional protein factors are necessary for the intron to be to be efficiently and accurately excised and the strands reattached. Group I introns require an external guanosine nucleotide as a cofactor. The 3'-OH of the guanosine nucleotide acts as a nucleophile to attack the 5'-phosphate of the intron's 5' nucleotide. The 3' end of the 5' exon is termed the splice donor site. The 3'-OH at the 3' splice donor end of the 5' exon next attacks the splice acceptor site at the 5' nucleotide of the 3' exon, releasing the intron and covalently attaching the two exons together.

23) Enlist the name of splicing methods?

There are two techniques in splicing of optical fibers depending on the insertion loss, cost, and performance characteristics. They are fusion splicing and mechanical splicing. The mechanical splicing is again divided into two types such as V-grooved splicing and elastic-tube splicing.

24) CRISPR stands for?

“CRISPR” (pronounced “crisper”) stands for Clustered Regularly Interspaced Short Palindromic Repeats, which are the hallmark of a bacterial defense system that forms the basis for CRISPR-Cas9 genome editing technology.

25) CRISPR applications?

DO NOT COPY

This is clearly the function of CRISPR RNAs, which have been demonstrated to repress bacteriophage and plasmid entry into the cell, and in principle could be used to silence genes from other mobile elements.

26) Name two proteins found in E.coli?

One, the *lac* repressor, acts as a lactose sensor.
The other, catabolite activator protein (CAP), acts as a glucose sensor.

27) Initiation complex?

Enhancers are sites on the DNA helix that are bound by activators in order to loop the DNA bringing a specific promoter to the initiation complex. Enhancers are much more common in eukaryotes than prokaryotes, where only a few examples exist.

28) What factors required for transcription in eukaryotes?

Eukaryotic transcription takes place in the nucleus of the cell and proceeds in three sequential stages: initiation, elongation, and termination. The transcriptional machinery that catalyzes this complex reaction has at its core three multi-subunit RNA polymerases. RNA polymerase I is responsible for transcribing RNA that codes for genes that become structural components of the ribosome. Protein coding genes are transcribed into messenger RNAs (mRNAs) that carry the information from DNA to the site of protein synthesis. Although mRNAs possess great diversity, they are not the most abundant RNA species made in the cell. The so called non- coding RNAs account for the large majority of the transcriptional output of a cell. These non- coding RNAs perform a variety of important cellular functions.

29) **Define operon model?**

The Operon Model of Gene Regulation. An operon is a cluster of genes that are transcribed together to give a single messenger RNA (mRNA) molecule, which therefore encodes multiple proteins. Such polycistronic mRNA is typically found in prokaryotes. Groups of genes coding for related proteins are arranged in units known as operons. An operonconsists of an operator, promoter, regulator, and structural genes. The regulator gene codes for a repressor protein that binds to the operator, obstructing the promoter (thus, transcription) of the structural genes.

30) Ctd tail?

The CTD extends from the core of the enzyme to form a tail-like structure that provides binding sites for various factors involving in RNA processing and transcription-coupled modifications. Having shape like a tail, that is called as CTD tail.

31) RNA editing?

RNA editing is a molecular process through which some cells can make discrete changes to specific nucleotide sequences within a RNA molecule after it has been generated by RNA

DO NOT COPY

polymerase. RNA editing is relatively rare, and common forms of RNA processing (e.g. splicing, 5'-capping and 3'-polyadenylation) are not usually included as editing. Editing events may include the insertion, deletion, and base substitution of nucleotides within the edited RNA molecule.

32) Two protein in E.Coli that are necessary for lactose metabolism?

More specifically, in E. coli, the switch from glucose use to lactose use depends on the presence of both cAMP and a molecule called catabolite activator protein (CAP).

33) Describe RNA processing event?

Elongating polymerase is associated with a set of protein factors required for various types of RNA processing. mRNA is capped as soon as it emerges from the RNA-exit channel of the polymerase. After capping, dephosphorylation of Ser-5 within the CTD repeats may be responsible for dissociation of the capping machinery. Further phosphorylation of Ser-2 causes recruitment of the RNA splicing machinery that catalyzes the removal of non-coding introns to generate mature mRNA. Alternative splicing expands the protein complements in eukaryotes. Just as with 5‘-capping and splicing, the CTD tail is involved in recruiting enzymes responsible for 3‘-polyadenylation, the final RNA processing event that is coupled with the termination of transcription.

34) Write five features of Operon Model?

1. The products of the lacZ, lacY and lacA genes are encoded in a single polycistronic lac mRNA molecule.
2. The promoter for this mRNA molecule is immediately adjacent to the lac region.
3. The operator is a sequence of bases (in the DNA) to which the repressor protein binds.

4. When the repressor protein is bound to the operator, lac mRNA transcription can’t take place.
5. Inducers stimulate lac mRNA synthesis by binding to the repressor. This binding alters the repressor’s conformation so it can’t bind to the operator. In the presence of an inducer, therefore, The operator is unoccupied and the promoter is available for initiation of mRNA synthesis. This state is called derepression.

35) **Involvement of CTD Tail In Transcription?**

RNA polymerase II transcription elongation and Pol II CTD Ser2 phosphorylation: Atail of two kinases. Author information: ... The transition between initiation and productive elongation during RNA Polymerase II (Pol II) transcription is a well-appreciated point of regulation across many eukaryotes.

36) How short life span of mRNA. Analyze two experiment? Write experiment Name and experiment explanation?

DO NOT COPY

The half life of a typical bacterial mRNA is a few minutes. The particular mRNA and their life span can be analyzed by the pulse-chase experiment. Bactria are briefly cultured in a medium that contains a radioactive precursor for RNA such as [^{3}H] uridine. Then the bacteria are switched to a medium that contains a high concentration of non radioactive uridine while no [^{3}H] uridine and samples are removed at specific times for analyses. The RNA are isolated and different types are separated by gel electrophoresis and detected by their radioactivity. A typical radioactive mRNA molecule will decrease with a half life of a few minutes, whereas radioactive rRNA and tRNA will remain through many generations. However, bacteria also contain some longlived mRNA. molecules which can not be analyzed by this technique.

1st Expriment Name: Pulse Chase

2nd Expreiment Pulse Chase: Pulse chase experiment, mRNA synthesized during a short time window was labelled. These studies showed that mRNA is synthesized in the nucleus. Within an hour, most of this RNA had left the nucleus and was observed in the end cytoplasm.

37) How Many CTD Heptatidase In Repeat Sequence In Yeast Drosophila And Human?

There are 27 of these repeats in the yeast Pol II CTD, 45 in the fly Drosophila, and 52 in humans.

38) What ADAR? 3 Genes close to Human ADAR?

Double-stranded RNA-specific adenosine deaminase is an enzyme that in humans is encoded by the **ADAR** gene (which stands for adenosine deaminase acting on RNA). ... The conversion from A to I in the RNA disrupt the normal A:U pairing that makes the RNA unstable.

Three Genes: Humans and mice contain three ADAR genes: ADAR1, ADAR2, and ADAR3.

39) RNA Elongation In Three Lines?

Elongation is the stage when the RNAstrand gets longer, thanks to the addition of new nucleotides. During elongation, RNA polymerase "walks" along one strand of DNA, known as the template strand, in the 3' to 5' direction.

40) How Regulatory Protein Control Gene Expression?

Regulated stages of gene expression. Any step of gene expression may be modulated, from the DNA- RNA transcription step to post-translational modification of a protein. The following is a list of stages where gene expression is regulated the most extensively utilised point is Transcription.

41) Write point operon?

DO NOT COPY

An operon is a functioning unit of DNA containing a cluster of genes under the control of a single promoter.

42) RNA polymerase I and III functions?

RNA polymerase I (RNAPI) transcribes rRNA genes, RNA polymerase II (RNAPII) transcribes mRNA, miRNA, snRNA, and snoRNA genes, and RNA polymerase III (RNAPIII) transcribes tRNA and 5S rRNA genes. This is in contrast with prokaryotes where a single RNA polymerase is responsible for the transcription of all genes.

43) Write requriemt For Transcription?

Important is the concept that transcription, whether prokaryotic or eukaryotic, has three main events. Initiation - binding of RNA polymerase to double-stranded DNA; this step involves a transition to single-strandedness in the region of binding; RNA polymerase binds at a sequence of DNA called the promoter.

44) Splicing?

Splicing is the editing of a newly made precursor messenger RNA (pre-mRNA) transcript into a mature messenger RNA (mRNA). After splicing, introns are removed and exons are joined together.

45) Gene Expression?

Gene expression is the process by which information from a gene is used in the synthesis of a functional gene product.

46) CRISPER and Application?

CRISPR technology is a simple yet powerful tool for editing genomes. It allows researchers to easily alter DNA sequences and modify gene function. Its many potential applications include correcting genetic defects, treating and preventing the spread of diseases and improving crops

47) **Plus chase experiment, to verify life short span of mRNA?**

Pulse chase" experiment, mRNA synthesized during a short time window was labelled. These studies showed that mRNA is synthesized in the nucleus. Within an hour, most of this RNA had left the nucleus and was observed in the end cytoplasm.

48) **Write The Process Of Elongating Polymerase?**

The Elongating Polymerase Is a processive machine that synthesizes and proofreads RNA. DNA passes through the elongating enzyme in a manner very similar to its passage through the open complex. Thus, doublestranded DNA enters the front of the enzyme between the pincers. At the

opening of the catalytic cleft, the strands separate to follow different paths through the enzyme. The strands exit via their respective channels and re-form a double helix **behind the elongating polymerase.**

49) Polarity Of DNA Helicase?

Polarityis ascertained by identifying the duplex region that is denatured by thehelicase. For helicases involved in DNA replication, the polarity of the reaction is strongly indicative of helicase placement on the leading (a 3 ' -5 'polarity) or lagging (a 5 ' +3 'polarity) strand.

50) Function Of CPD Photolyase?

Increase in CPD photolyase activity functions effectively to prevent growth inhibition caused by UVB radiation. ... Conversely, the AS-D plant had little photolyase activity, was severely damaged by elevated UVB radiation, and maintained higher CPD levels in its leaves during growth under UVB radiation.

51) Advancement Of Molecular Bio In Agriculture Science?

Agricultural moelcular is a collection of scientific techniques used to improve plants, animals and microorganisms. Based on an structure and characteristics of DNA, scientists have developed solutions to increase agricultural productivity. Scientists have learned how to move genes from one organism to another. This has been called genetic modification (GM), genetic engineering (GE) or genetic improvement (GI).

52) Standrd Amino Acids 5 Name?

Histidine, Isoleucine, Leucine, Lysine,Methionine, Phenylalanine, Threonine, Tryptophan, And Valine

53) 2 Mutations Which Cause One Base Change?

The simplest mutations are switches of one base for another. There are two kinds of such mutations which include:-

- Transitions
- Transversions

54) Types Of UV Light?

Ultraviolet light is divided into three bands: UV –A (321-400nm) UV –B (296-320nm) UV –C (100-295nm).

55) Role Of Regularory Proteins In Gene Expression?

DO NOT COPY

Genes are very often controlled by extracellular signals; in the case of bacteria, these signals typically mean molecules present in the growth medium. These signals are communicated to genes by regulatory proteins, which come in two types: positive regulators, or activators, and negative regulators, or repressors. Typically, these regulators are DNAbinding proteins that recognize specific sites at or near the genes they control. An activator increases. transcription of the regulated gene, and repressors decrease or eliminate that transcription.

56) Splicing and Classes?

Splicing is the editing of a newly made precursor messenger RNA Exons, introns, alternative splicing.

57) Lactose metabolism?

Metabolism: Your body starts metabolizing or breaking down lactose in the intestines with the help of lactase, an enzyme produced and released by cells that line your small intestine. The enzyme attaches to lactose and cleaves it into molecules of galactose and glucose

58) Transition work with open complex?

Transition to the Open Complex involves structural changes in RNA Polymerase and in the Promoter DNA. The initial binding of. RNA polymerase to the promoter DNA in the closed complex leaves the DNA in double-stranded form. The transition from the closed to the open complex involves structural changes in the enzyme and the opening of the DNA double helix to reveal the template and nontemplate strand

59) Half Life of mRNA?

In prokaryotes, which lack a cell nucleus, transcription and translation occur nearly simultaneously. Prokaryotic mRNAs have very short half-lives (with an averagehalf-life of 3 minutes) and are usually rapidly degraded by exonucleases that start at the 3' end and move **towards the 5' end.**

60) What are simple lipids? Give two types of simple lipids?

A simple lipid is a fatty acid ester of different alcohols and carriers no other substances. These lipids belong to a heterogeneous class of predominantly nonpolar compounds, mostly insoluble in water, but soluble in nonpolar organic solvents such as chloroform and benzene.

Example: fats and oils.

61) Name three functional group of purines and pyrimidines?

Nitro group, exocyclic amino group and carbonyl group.

62) Two example of unnatural pyrimidine bases?

DO NOT COPY

Fluorouracil or Azc

63) Differentiate b/w the structure of the thymine and uracil?

The only difference between thymine and uracil is a methyl group- thymine has it but uracil doesn't.

64) Name the pentose sugars present in nucleotide. Also write down its characteristics?

Ribose and deoxyribose. The 5-carbon sugar ribose and deoxyribose are important components of nucleotide and are found in RNA and DNA. In chemistry, a glycosidic bond or glycosidic linkage us a type of. Covalent bond that joins a carbohydrate molecule to another group, which may or may not be another carbohydrate. Cyclic guanosine monophosphate (cGMP) is a cyclic nucleotide derived from guanosine triphosphate (GTP). cGMP acts as a second messenger much like cyclic AMP. Its most likely mechanism of action is activation of intracellular protein kinases in response to the binding of membrane -impermeable peptide hormones to the external surface.

65) Splicing methods?

There are 2 methods of splicing, mechanical or fusion. Both methods provide much lower insertion loss compared to fiber connectors.

66) Galactose operon?

The gal operon is a prokaryotic operon, which encodes enzymes necessary for galactose metabolism. The operon contains two operators, O_E (for external) and O_I. The former is just before the promoter, and the latter is just after the galE gene (the first gene in the operon). The gal operon of E. coli consists of 3 structural genes: galE (epimerase), galT (galactose transferase), and galK (galactokinase), which are transcribed from two overlapping promoters PG1 and PG2 upstream from galE.

67) Transcription step elongation?

Promoters can differ in "strength"; that is, how actively they promote transcription of their adjacent DNA sequence. Promoter strength is in many (but not all) cases, a matter of how tightly RNA polymerase and its associated accessory proteins bind to their respective DNA sequences. The more similar the sequences are to a consensus sequence, the stronger the binding is. Additional transcription regulation comes from transcription factors that can affect the stability of the holoenzyme structure at initiation. Most transcripts originate using adenosine-5'-triphosphate (ATP) and, to a lesser extent, guanosine-5'-triphosphate (GTP) (purine nucleoside triphosphates) at the +1 site. Uridine-5'- triphosphate (UTP) and cytidine-5'-triphosphate (CTP) (pyrimidine nucleoside triphosphates) are disfavoured at the initiation site.

68) Central dogma?

DO NOT COPY

The ‗Central Dogma' is the process by which the instructions in DNA are converted into a functional product. It was first proposed in 1958 by Francis Crick, discoverer of the structure of DNA.

The central dogma states that the pattern of information that occurs most frequently in our cells is:

- From existing DNA to make new DNA
- From DNA to make new RNA (transcription)
- From RNA to make new proteins (translation).

69) What happened in cellular machinery and function if pyrimidine dimer is not removed?

The (6-4) photoproduct causes a major distortion in B-DNA because the two pyrimidine rings are perpendicular to each other. If not removed, a pyrimidine dimer or (6-4) photoproduct can interfere with the normal operation of the replication and transcription machinery. This interference results in mutations and cell death. Even if the lesion is removed, the result can be a mutation.

LONG QUESTIONS

70) Base repair excision system

Base Excision Repair

Many types of DNA damages can't be repaired by a single enzyme that catalyses direct damage reversal. Instead, repair requires the participation of several different enzymes, each performing a specific task in multistep pathway. Damage to DNA bases caused by deamination, oxidation, and alkylation is mainly repaired by one such multistep pathway which is called, Base Excision Repair (BER). BER pathway is same in all organisms. Enzymes involved in the base excision repair pathway also participate in the repair of single-strand break in DNA. Base excision repair derives its name from the first step in the pathway, N-glycosyl bond cleavage. This cleavage excises the damaged or inappropriate base from the DNA to form a basic site. Because no single enzyme can distinguish the four bases present in DNA from a wide variety of altered bases, cells must use many different enzymes to perform this function. Some N- glycosylases are monofunctional enzymes with only the ability to excise a damaged base. Such enzymes are called DNA glycosylases. Other N-glycosylases have an AP lyase activity that cleaves the bond between sugar and the phosphate 3' to the damaged site. This enzymes is designated as DNA glycosylase/lyase. Both types of enzymes detect the damaged base, flip it out of the DNA helix into an active site pocket, and then cleave the N-glycosyl bond. Although some glycosylases excise a specific base, most have a somewhat broader specificity. The E.coli enzyme Uracil N-glycosylase (Ung) is specific for uracil. Whereas another E. coli N-glycosylase, 3-methyladenine DNA glycosylase (AlkA), acts on 3- or 7-methyl purines, 3- or 7-ethylpurine, ethenoadenine, and O 2 –methyl pyrimidine. A null mutation in a gene that codes for a DNA glycosylase or

DO NOT COPY

DNA glycosylase/lyase is not lethal which probably reflects overlapping functional abilities among the glycosylases.

71) RNA editing

RNA editing

RNA editing is a molecular process through which some cells can make discrete changes to specific nucleotide sequences within a RNA molecule after it has been generated by RNA polymerase. RNA editing is relatively rare, and common forms of RNA processing (e.g. splicing, 5'-capping and 3'-polyadenylation) are not usually included as editing. Editing events may include the insertion, deletion, and base substitution of nucleotides within the edited RNA molecule. RNA editing has been observed in some tRNA, rRNA, mRNA or miRNA molecules of eukaryotes and their viruses, archaea and prokaryotes. RNA editing occurs in the cell nucleus and cytosol, as well as within mitochondria and plastids. In vertebrates, editing is rare and usually consists of a small number of changes to the sequence of affected molecules. In other organisms, extensive editing (pan-editing) can occur; in some cases the majority of nucleotides in an mRNA sequence may result from editing. RNA-editing processes show great molecular diversity, and some appear to be evolutionarily recent acquisitions that arose independently. The diversity of RNA editing phenomena includes nucleobase modifications such as cytidine (C) to uridine (U) and adenosine (A) to inosine (I) deamination, as well as non-templated nucleotide additions and insertions. RNA editing in mRNAs effectively alters the amino acid sequence of the encoded protein so that it differs from that predicted by the genomic DNA sequence. Editing by insertion or deletion RNA editing through the addition and deletion of uracil has been found in kinetoplasts from the mitochondria of Trypanosoma brucei[3] Because this may involve a large fraction of the sites in a gene, it is sometimes called "pan-editing" to distinguish it from topical editing of one or a few sites. Pan-editing starts with the base-pairing of the unedited primary transcript with a guide RNA (gRNA), which contains complementary sequences to the regions around the insertion/deletion points. The newly formed double-stranded region is then enveloped by an editosome, a large multi-protein complex that catalyses the editing. The editosome opens the transcript at the first mismatched nucleotide and starts inserting uridines. The inserted uridines will base-pair with the guide RNA, and insertion will continue as long as A or G is present in the guide RNA and will stop when a C or U is encountered. The inserted nucleotides cause a frameshift and result in a translated protein that differs from its gene. The mechanism of the editosome involves an endonucleolytic cut at the mismatch point between the guide RNA and the unedited transcript. The next step is catalysed by one of the enzymes in the complex, a terminal U-transferase, which adds us from UTP at the 3‘end of the mRNA. The opened ends are held in place by other proteins in the complex. Another enzyme, a U-specific exoribonuclease, removes the unpaired us. After editing has made mRNA complementary to gRNA, an RNA ligase re-joins the ends of the edited mRNA transcript. As a consequence, the editosome can edit only in a 3‘to 5‘direction along the primary RNA transcript. The complex can act on only a single guide RNA at a time. Therefore, a RNA transcript requiring extensive editing will need more than one guide RNA and editosome complex.

DO NOT COPY

72) RNA processing in detail

RNA processing

Elongating polymerase is associated with a set of protein factors required for various types of RNA processing. mRNA is capped as soon as it emerges from the RNA-exit channel of the polymerase. After capping, dephosphorylation of Ser-5 within the CTD repeats may be responsible for dissociation of the capping machinery. Further phosphorylation of Ser-2 causes recruitment of the RNA splicing machinery that catalyses the removal of non-coding introns to generate mature mRNA. Alternative splicing expands the protein complements in eukaryotes. Just as with 5‘-capping and splicing, the CTD tail is involved in recruiting enzymes responsible for 3‘-polyadenylation, the final RNA processing event that is coupled with the termination of transcription.

73) Spliceosome

Spliceosome

The spliceosome has been described as one of "the most complex macromolecular machines known," "composed of as many as 300 distinct proteins and five RNAs" (Nilsen, 2003). The animation above reveals this astonishing machine at work on the precursor mRNA, cutting out the non-coding introns and splicing together the protein-coding exons. A spliceosome is a large and complex molecular machine found primarily within the splicing speckles of the cell nucleus of eukaryotic cells. The spliceosome is assembled from snRNAs and protein complexes. The spliceosome removes introns from a transcribed pre-mRNA, a kind of primary transcript. This process is generally referred to as splicing. Only eukaryotes have spliceosomes and metazoans have a second spliceosome, the minor spliceosome. Introns (which, unlike exons, do not code for proteins) can be of considerable length in higher eukaryotes, even spanning many thousands of bases and sometimes comprising some 90% of the precursor mRNA. In contrast, lower eukaryotes such as yeast possess fewer and shorter introns, which are typically fewer than 300 bases in length. Since introns are the non-coding segments of genes, they are removed from the mRNA before it is translated into a protein. This is not to say, of course, that introns are without important function in the cell (as I discuss here). Comprising the spliceosome, shown at right (excerpted from Frankenstein et al., 2012) are several small nuclear ribonucleoproteins (snRNPs) -- called U1, U2, U4, U5 and U6 -- each of which contains an RNA known as an snRNA (typically 100-300 nucleotides in length) -- and many other proteins that each contribute to the process of splicing by recognizing sequences in the mRNA or promoting rearrangements in spliceosome conformation. The spliceosome catalyses a reaction that results in intron removal and the "gluing" together of the protein-coding exons. RNA Splicing The first stage in RNA splicing is recognition by the spliceosome of splice sites between introns and exons. Key to this process are short sequence motifs. These include the 5' and 3' splice sites (typically a GU and AG sequence respectively); the branch point sequence (which contains a conserved adenosine

DO NOT COPY

important to intron removal); and the polypyrimidine tract (which is thought to recruit factors to the branch point sequence and 3' splice site).

74) How group-1 splicing work in DNA

Group – 1

Group I introns occur in nuclear, mitochondrial and chloroplast rRNA genes, group II introns in mitochondrial and chloroplast mRNA genes. Many of the group I and group II introns are self-splicing in that no additional protein factors are necessary for the intron to be efficiently and accurately excised and the strands reattached. "The nucleotide sequence of group II self-splicing introns is highly conserved, and hence these introns fold into an evolutionarily conserved three-dimensional structure, which can undergo a self-splicing reaction in the absence of any trans-acting factors. In contrast, the nucleotide sequences and length of nuclear pre-mRNA introns is highly variable, except for the short conserved sequences at the 5´ and 3´ splice sites and the branch points. Therefore nuclear pre-mRNA splicing requires trans-acting factors, which interact with these short conserved sequences, and from which the catalytically active spliceosome is assembled. The conserved sequences are: 5' splice site = AGguragu; 3' splice site = yyyyyyy nagG (y= pyrimidine); branch site = ynyuray (r = purine, n = nucleotide) Expressed differently, the highly conserved, consensus sequence for the 5' donor splice site is (for RNA): (A or C)AG/GUAAGU. That is, most exons end with AG and introns begin with GU (GT for DNA). The highly conserved, consensus sequence for the 3' acceptor splice site is (for RNA): (C/U) less than 10N(C/T) AG/G, where most introns end in AG after a long stretch of pyrimidines. The branch site within introns (area of lariat formation close to the acceptor site during splicing) has the consensus sequence UAUAAC. In most cases, U can be replaced by C and A can be replaced by G. However, the penultimate (bold) a residue is fully conserved (invariant). Group I introns require an external guanosine nucleotide as a cofactor. The 3'-OH of the guanosine nucleotide acts as a nucleophile to attack the 5'-phosphate of the intron's 5' nucleotide. The 3' end of the 5' exon is termed the splice donor site. The 3'-OH at the 3' splice donor end of the 5' exon next attacks the splice acceptor site at the 5' nucleotide of the 3' exon, releasing the intron and covalently attaching the two exons together. Pre-mRNA processing takes place in the nucleus of eukaryotes, whereas lack of a nuclear membrane in prokaryotes permits initiation of translation while transcription is not yet complete. Pre-mRNA processing events include capping of the 5‘end on the pre-mRNA, pre-mRNA splicing to remove intronic sequences, and polyadenylation of the 3‘end of the pre-mRNA.

75) Lac operon model and its five points

Lac operon

Lac operon (lactose operon) is an operon required for the transport and metabolism of lactose in Escherichia coli and many other enteric bacteria. Although glucose is the preferred carbon source

for most bacteria, the lac operon allows for the effective digestion of lactose when glucose is not available. Gene regulation of the lac operon was the first genetic regulatory mechanism to be understood clearly, so it has become a foremost example of prokaryotic gene regulation. It is often discussed in introductory molecular and cellular biology classes at universities for this reason. Bacterial operons are polycistronic transcripts that are able to produce multiple proteins from one mRNA transcript. In this case, when lactose is required as a sugar source for the bacterium, the three genes of the lac operon can be expressed and their subsequent proteins translated: lacZ, lacY, and lacA. The gene product of lacZ is β-galactosidase which cleaves lactose, a disaccharide, into glucose and galactose. LacY encodes lactose permease, a protein which becomes embedded in the cytoplasmic membrane to enable transport of lactose into the cell. Finally, lacA encodes galactoside O-acetyltransferase. It would be wasteful to produce the enzymes when there is no lactose available or if there is a more preferable energy source available, such as glucose. The lac operon uses a two-part control mechanism to ensure that the cell expends energy producing the enzymes encoded by the lac operon only when necessary. In the absence of lactose, the lac repressor halts production of the enzymes encoded by the lac operon. In the presence of glucose, the catabolite activator protein (CAP), required for production of the enzymes, remains inactive, and EIIAGlc shuts down lactose permease to prevent transport of lactose into the cell. This dual control mechanism causes the sequential utilization of glucose and lactose in two distinct growth phases, known as diauxie. The lac operon consists of three structural genes, and a promoter, a terminator, regulator, and an operator. The three structural genes are: lacZ, lacY, and lacA. = lacZ encodes β-galactosidase (LacZ), an intracellular enzyme that cleaves the disaccharide lactose into glucose and galactose. lacY encodes lactose permease (LacY), a transmembrane symporter that pumps βgalactosides into the cell using a proton gradient in the same direction. lacA encodes galactoside O-acetyltransferase (LacA), an enzyme that transfers an acetyl group from acetyl-CoA to β-galactosides.

Points

The five major features of the model are:-

1. The products of the lacZ, lacY and lacAgenes are encoded in a single polycistroniclac mRNA molecule.

2. The promoter for this mRNA molecule is immediately adjacent to the lac region. Promoter mutations (p-) that are completely incapable of making β-galactosidase, permease, and transacetylase have been isolated. The promoter is located between lacI and lacO.

3. The operator is a sequence of bases (in the DNA) to which the repressor protein binds.

4. When the repressor protein is bound to the operator, lac mRNA transcription can't take place.

5. Inducers stimulate lac mRNA synthesis by binding to the repressor. This binding alters the repressor's conformation so it can't bind to the operator. In the presence of an inducer, therefore, the operator is unoccupied and the promoter is available for initiation of mRNA synthesis. This state is called derepression.

DO NOT COPY

76) Five points of operon model

5 points

The five major features of the model are:-

- The products of the lacZ, lacY and lacAgenes are encoded in a single polycistroniclac mRNA molecule.
- The promoter for this mRNA molecule is immediately adjacent to the lac region. Promoter mutations (p-) that are completely incapable of making β-galactosidase, permease, and transacetylase have been isolated. The promoter is located between lacI and lacO.
- The operator is a sequence of bases (in the DNA) to which the repressor protein binds.
- When the repressor protein is bound to the operator, lac mRNA transcription can't take place.
- Inducers stimulate lac mRNA synthesis by binding to the repressor. This binding alters the repressor's conformation so it can't bind to the operator. In the presence of an inducer, Therefore, the operator is unoccupied and the promoter is available for initiation of mRNA synthesis. This state is called derepression.

77) What is CRISPRs Work, application

Clustered regularly interspaced short palindromic repeats (CRISPR, pronounced crisper) are segments of prokaryotic DNA containing short, repetitive base sequences. In a palindromic repeat, the sequence of nucleotides is the same in both directions. Each repetition is followed by short segments of spacer DNA from previous exposures to foreign DNA (e.g., a virus or plasmid). Small clusters of CAS (CRISPR-associated system) genes are located next to CRISPR sequences. The CRISPR/Cas system is a prokaryotic immune system that confers resistance to foreign genetic elements such as those present within plasmids and phages that provides a form of acquired immunity. RNA covering the spacer sequence helps Cas proteins recognize and cut

Exogenous DNA. Other RNA-guided Cas proteins cut foreign RNA. CRISPRs are found in approximately 40% of sequenced bacterial genomes and 90% of sequenced archaea. A simple version of the CRISPR/Cas system, CRISPR/Cas9, has been modified to edit genomes. By delivering the Cas9 nuclease complexed with a synthetic guide RNA (gRNA) into a cell, the cell's genome can be cut at a desired location, allowing existing genes to be removed and/or new ones added. The Cas9-gRNA complex corresponds with the CAS III crRNA complex in the above diagram. CRISPR/Cas genome editing techniques have many potential applications, including medicine and crop seed enhancement. The use of CRISPR/Cas9-gRNA complex for genome editing was the AAAS's choice for breakthrough of the year in 2015. Bioethical concerns have been raised about the prospect of using CRISPR for germline editing.

Application

DO NOT COPY

- Genome engineering
- Knockdown/activation
- RNA editing
- Disease models
- Gene drive
- Biomedicine
- Gene function
- In vitro genetic depletion
- Patents and commercialization

CRISPR stands for

Clustered Regularly Interspaced Short Palindromic Repeats (or CRISPRs).

78) Gene expression controlled by regulatory proteins.

Genes are very often controlled by extracellular signals; in the case of bacteria, these signals typically mean molecules present in the growth medium. These signals are communicated to genes by regulatory proteins, which come in two types positive regulators, or activators, and negative regulators, or repressors. Typically, these regulators are DNA-binding proteins that recognize specific sites at or near the genes they control. An activator increases transcription of the regulated gene, and repressors decrease or eliminate that transcription.

The way these proteins regulate gene expression

First, RNA polymerase binds to the promoter in a closed complex (in which the DNA strands remain together). The polymerase–promoter complex then undergoes a transition to an open complex in which the DNA at the start site of transcription is unwound and the polymerase is positioned to initiate transcription. This is followed by promoter escape, the step in which polymerase leaves the promoter and starts transcribing. Polymerase then proceeds through the elongation phase before finally terminating. Which steps are stimulated by activators and inhibited by repressors depends on the promoter. The most common step at which gene expression is regulated is the initiation of transcription.

There are two reasons why this might make sense.

First, transcription initiation is the most energetically efficient step to regulate. By this we mean that deciding whether or not to express a gene at the first step ensures that no energy or resources are wasted.

Second, regulation at this first step is easier to do.

There is only a single copy of each gene and so typically only a single promoter on a single DNA molecule must be regulated to control expression of a given gene. Regulating later steps can have two advantages. First, it allows for more inputs: if a gene is regulated at more than one step, more signals can modulate its expression, or the same signals can do so even more effectively. Second, regulation at steps later than transcription initiation can reduce the response time. Thus,

DO NOT COPY

consider again the example of translational regulation. If a signal relieves repression of this step, the protein product encoded by the gene will be produced immediately upon receipt of that signal. This reduced response time might obviously be advantageous in some situations. Many promoters are regulated by activators that help RNA polymerase bind DNA and by repressors that block that binding. At many promoters, in the absence of regulatory proteins, RNA polymerase binds only weakly. This is because one or more of the promoter elements is absent or imperfect. When polymerase does occasionally bind, however, it spontaneously undergoes a transition to the open complex and initiates transcription. This gives a low level of constitutive expression called the basal level. Binding of RNA polymerase is the rate-limiting step in this case. To control expression from such a promoter, a repressor need only bind to a site overlapping the region bound by polymerase. In that way, the repressor blocks polymerase binding to the promoter, thereby preventing transcription, although it is important to note that repression can work in other ways as well. The site on DNA where a repressor binds is called an operator. To activate transcription from this promoter, an activator can just help the polymerase bind the promoter.

79) RNA splicing

RNA splicing

RNA splicing:

The coding sequence of a gene is a series of three nucleotide codons that specifies the linear sequence of amino acids in its polypeptide product. It is generally assumed that the coding sequence is contiguous; i,e., the codon for one amino acid is immediately adjacent to the codon for the next amino acid in the polypeptide chain. This is true in the vast majority of cases in bacteria and their phage. But it is rarely so for eukaryotic genes. In those cases, the coding sequence is interrupted by stretches of non-coding sequences. Many eukaryotic genes are thus mosaics, consisting of blocks of coding sequences separated from each other by blocks of non-coding sequences. The coding sequences are called exons and the intervening sequences are called introns. Once transcribed into an RNA transcript, the introns must be removed and the exons joined together to create the mRNA for that gene. The number of introns found within a gene varies enormously-from one in the case of most yeast genes (and a few human genes), to 50 in the case of the chicken pro. 2 collagen gene to as many as 363 in the case of the Titin gene of humans. The sizes of the exons and introns vary as well. Indeed, introns are very often much longer than the exons they separate. Thus, for example, exons are typically on the order of 150 nucleotides, whereas introns— although they too can be short—can be as long as 800,000 nucleotides (800 kb). As another example, the mammalian gene for the enzyme dihydrofolatereductase is more than 31 kb long, and within it are dispersed six exons that correspond to 2 kb of mRNA. Thus, in this case, the coding portion of the gene is,10% of its total length. Like the uninterrupted genes of prokaryotes, the split genes of eukaryotes are transcribed into a single RNA copy of the entire gene - the primary transcript that contains introns as well as exons. Because the length and number of introns, the primary transcript (or pre-mRNA) can be very

DO NOT COPY

long indeed. The process of intron removal is called RNA Splicing. It converts the pre-mRNA into mature mRNA containing only exons. RNA Splicing must occur with great precision to avoid the loss, or addition, of even a single nucleotide at the sites at which the exons are joined. The triplet-nucleotide codons of mRNA are translated in a fixed reading frame that is set by the first codon. Lack of precision in splicing will change the reading frames of exons. Some pre-mRNAs can be spliced in more than one way. Thus, mRNAs containing different selections of exons can be generated from a given pre- mRNA Alternative splicing strategy enables a gene to give rise to more than one polypeptide product. These alternative products are called isoforms. It is estimated that 90% or more of the protein-coding genes in the human genome are spliced in alternative ways to generate more than one isoform.

80) Describe RNA processing event

Initiator tRNA (containing methionine) binds to small ribosomal subunit at the P site along with translation initiation factors. unit binds to 5' end of mRNA looking for AUG, initiation factors dissociate when AUG is found large ribosomal subunit assembles. tRNA with next amino acid binds to the large ribosomal subunit at the A site. Initiator tRNA separated from methionine at P site by breaking of energy bonds. C terminal of methionine binds to N terminal of new amino acid using enzyme peptidyl transferase (part of the ribosome). large subunit moves relative to the small subunit (which is still reading the mRNA) so that the tRNA with the newly bound amino acid shifts to the P site small subunit moves 3 nucleotides along mRNA to read next codon. New tRNA carrying the next amino acid is bound to a site and process from step 4 to 8 repeats. When small ribosomal
Subunit reads stop codons UAA, UAG or UGA, release factors bind to the A site. Release Factors alter the activity of peptidyl transferase so that water is catalysed instead of peptide bond. C terminal of newly-formed protein is released into the cytoplasm where it is folded with, or without, the help of chaperone proteins Some simple notes about translation:

- mRNA always complementary to template strand of DNA and same sequence as coding strand of DNA save for switching T for U. It is read from 5' to 3'.
- tRNA anticodons are complementary to mRNA codons, same as the template strand DNA
- nucleotides save for switching T for U, and is complementary to the coding strand of DNA

Since you just asked about RNA processing, I'm assuming that you don't need me to go over transcription in detail as well. Essentially, transcription is just RNA polymerase making mRNA that is complementary to the template strand of the DNA. The mRNA has introns that are spiced out, then are capped and added with a poly-A tail. The final product is what gets through the nuclear pores and into the ribosome.

81) Work of Claud S.Ruperd and his co-worker.

Claud S. Rupert and co-workers devised an in vitro photo reactivation system in 1957, taking a major step toward determining the chemical mechanism of photo reactivation. They used a

DO NOT COPY

straightforward approach in which they isolated DNA from the gram-negative bacteria Haemophilus influenza. They irradiated this DNA with UV light to inactivate its transforming ability. Then they demonstrated that a cell-free E. coli extract restored the transforming activity in the presence of light. Although, there study had the potential to open the way for the purification and characterization of photo reactivation enzyme, investigators still needed to establish the chemical nature of this repair.

82) Mechanism of DNA polymerase

DNA polymerase:

The DNA polymerases are enzymes that create DNA molecules by assembling nucleotides, the building blocks of DNA. These enzymes are essential to DNA replication and usually work in pairs to create two identical DNA strands from a single original DNA molecule. Every time a cell divides, DNA polymerase is required to help duplicate the cell‘s DNA, so that a copy of the original DNA molecule can be passed to each of the daughter cells. It uses a single active site to catalyse the addition of any of four deoxy nucleoside triphosphates. DNA polymerase monitors the ability of the incoming nucleotide to form an A: T or G: C base pair, rather than detecting the exact nucleotide that enters the active site. Only when a correct nucleotide comes, the 3‘-OH of the primer and the α-phosphate of the nucleotide align in optimum position for catalysis to take place. Incorrect base pairing leads to dramatically lower rate of nucleotide addition as a result of catalytically unfavourable alignment of these substrates. DNA polymerase shows an impressive ability to distinguish between ribonucleoside (rNTPs) and deoxyribonucleoside triphosphates (dNTPs). Although rNTPs are present at approx. ten-fold higher concentration in the cell, yet their incorporation rate is 1000-folds lower than dNTPs. This discrimination is mediated by the steric exclusion of rNTPs from the active site of DNA polymerase. In DNA polymerase, the nucleotide-binding pocket cannot accommodate a 2‘-OH on the in-coming nucleotide. This space is occupied by two amino acids that make van der Waals contacts with the deoxyribose ring. These amino acids are called discriminator amino acids.

83) Gal operon/ Galactose operon

Gal operon the gal operon is a prokaryotic operon, which encodes enzymes necessary for galactose metabolism. The operon contains two operators, OE (for external) and OI. The former is just before the promoter, and the latter is just after the galE gene (the first gene in the operon). Repression of gene expression works via binding of repressor molecules to the two operators. These repressors dimerize, creating a loop in the DNA. The loop as well as hindrance from the external operator prevent RNA polymerase from binding to the promoter, and thus prevent transcription. The gal operon of E. coli consists of 3 structural genes: galE (epimerase), galT (galactose transferase), and galK (galactokinase), which are transcribed from two overlapping promoters PG1 and PG2 upstream from galE. Regulation of the operon is complex since the GalE product, an epimerase that converts UDP-glucose into UDP-galactose, is required for the formation of UDP-galactose for cell wall biosynthesis, in particular the cell wall component lipopolysaccharide, even when cells are not using galactose as a carbon/energy source. The gal operon is controlled by CRP-cAMP as for the lac operon. CRP-cAMP binds to -35 region promoting transcription from PG1 but inhibiting transcription from PG2. When cells are grown in glucose, basal level transcription occurs from PG2. The unlinked galR gene encodes the repressor for this system. A tetrameric GalR repressor binds to 2 operators, one located at +55

and one located at -60 relative to the PG1 start site. Looping of the DNA blocks the access of RNA polymerase to promoters and/or inhibits formation of the open complex. When GalR binds as a dimer to the -60 site only, promoter PG2 is activated, not repressed, allowing basal levels of GalE to be produced. In this state promoter PG1 is inactivated through interactions with the alpha subunit of RNA polymerase.

84) Elongation step of transcription

Elongation Promoters can differ in "strength"; that is, how actively they promote transcription of their adjacent DNA sequence. Promoter strength is in many (but not all) cases, a matter of how tightly RNA polymerase and its associated accessory proteins bind to their respective DNA sequences. The more similar the sequences are to a consensus sequence, the stronger the binding is. Additional transcription regulation comes from transcription factors that can affect the stability of the holoenzyme structure at initiation. Most transcripts originate using adenosine-5'-triphosphate (ATP) and, to a lesser extent, guanosine-5'-triphosphate (GTP) (purine nucleoside triphosphates) at the +1 site. Uridine-5'- triphosphate (UTP) and cytidine-5'-triphosphate (CTP) (pyrimidine nucleoside triphosphates) are disfavoured at the initiation site.

85) Eukaryotic transcription factors

Pol II-transcribed genes contain a region in the immediate vicinity of the transcription start site (TSS) that binds and positions the preinitiation complex. This region is called the core promoter because of its essential role in transcription initiation. Different classes of sequence elements are found in the promoters. For example, the TATA box is the highly conserved DNA recognition sequence for the TATA box binding protein, TBP, whose binding initiates transcription complex assembly at many genes. Eukaryotic genes also contain regulatory sequences beyond the core promoter. These cisacting control elements bind transcriptional activators or repressors to increase or decrease transcription from the core promoter. Well-characterized regulatory elements include enhancers, silencers, and insulators. These regulatory sequences can be spread over a large genomic distance, sometimes located hundreds of kilo bases from the core promoters. General transcription factors are a group of proteins involved in transcription initiation and regulation. These factors typically have DNA-binding domains that bind specific sequence elements of the core promoter and help recruit RNA polymerase to the transcriptional start site. General transcription factors for RNA polymerase II include TFIID, TFIIA, TFIIB, TFIIF, TFIIE, and TFIIH.

DO NOT COPY

Separate Whatsapp Groups for Girls and Boys

Special Internship Group with Senior Guideline

Zero Percentage Irrelevant Chatting in study groups

Follow this Link for Joining Study Group:

https://chat.whatsapp.com/D2yLdI6LFFpJuS2EKWYmbE

Remember Us in Your Prayers

DO NOT COPY